

Homework 7: Solutions

Problem 1.

Construct the adjoint representation of $SU(2)$ (antihermitian 3×3 matrices) in two ways:

- a) Using the structure constants $f_{ab}{}^c = \epsilon_{abc}$, and
- b) Using raising (J_+) and lowering (J_-) operators. (In this case J_+ , J_- and J_3 generate $SU(1,1)$ which is the Cartan-Weyl basis for $SU(2)$. So you have to transform from hermitian J_+ and J_- to antihermitian T_1 and T_2 .)

Find the similarity transformation S which maps $T_a^{(II)}$ obtained in the second case to the $T_a^{(I)}$ of the first case

$$ST_a^{(II)}S^{-1} = T_a^{(I)}. \quad (1)$$

Solution: In this problem we use some $SU(2)$ technology which we assume to be known. We adopt the following conventions

Compact basis: $[T_a, T_b] = \epsilon_{abc}T_c$; $T_a^\dagger = -T_a$, 3×3 matrices

CW basis: $J_+ = J_1 + iJ_2$, $J_- = J_1 - iJ_2$; $-iJ_a = T_a$ (2)

$$J_+|j, m\rangle = \sqrt{(j+m+1)(j-m)}|j, m+1\rangle; \quad J_-|j, m\rangle = \sqrt{(j+m)(j-m+1)}|j, m-1\rangle.$$

- a) Using $(T_a)^b{}_c = f_{ac}{}^b = \epsilon_{acb}$ (one may check that they satisfy $[T_a, T_b] = \epsilon_{abc}T_c$), we get

$$T_1^{(I)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}; \quad T_2^{(I)} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}; \quad T_3^{(I)} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (3)$$

One may check that $[T_1, T_2] = T_3$.

- b) $J_+|1, 1\rangle = 0$, $J_+|1, 0\rangle = \sqrt{2}|1, 1\rangle$, $J_+|1, -1\rangle = \sqrt{2}|1, 0\rangle$. So

$$J_+ = \sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4)$$

Then

$$J_- = J_+^\dagger = \sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}; \quad J_3 = \frac{1}{2}[J_+, J_-] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (5)$$

(One can also determine J_- and J_3 in the way we did for J_+ .) Hence

$$J_1 = \frac{J_+ + J_-}{2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}; \quad J_2 = \frac{J_+ - J_-}{2i} = \frac{1}{i\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}; \quad J_3 = \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}. \quad (6)$$

Then $T_a^{(II)} = -iJ_a^{(II)}$, hence

$$T_1^{(II)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ -i & 0 & -i \\ 0 & -i & 0 \end{pmatrix}; \quad T_2^{(II)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}; \quad T_3^{(II)} = \begin{pmatrix} -i & & \\ & 0 & \\ & & i \end{pmatrix}. \quad (7)$$

We must now find a matrix S such that $ST_a^{(II)} = T_a^{(I)}S$. By brute force (maybe a simpler method exists), we find

$$\begin{aligned} S_{12}(-i) &= 0, \quad \frac{1}{\sqrt{2}}S_{22}(-i) = -S_{31}, \quad S_{11}(-i) + S_{13}(-i) = 0, \quad \text{etc.} \\ \Rightarrow S &= \begin{pmatrix} -i & 0 & i \\ 1 & 0 & 1 \\ 0 & i\sqrt{2} & 0 \end{pmatrix}. \end{aligned} \quad (8)$$

As a check we evaluate

$$\begin{aligned} ST_1^{(II)} &= T_1^{(I)}S : \begin{pmatrix} -i & 0 & i \\ 1 & 0 & 1 \\ 0 & i\sqrt{2} & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ -i & 0 & -i \\ 0 & -i & 0 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} -i & 0 & i \\ 1 & 0 & 1 \\ 0 & i\sqrt{2} & 0 \end{pmatrix} \\ &\quad \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -2i & 0 \\ \sqrt{2} & 0 & \sqrt{2} \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -i\sqrt{2} & 0 \\ 1 & 0 & 1 \end{pmatrix} \text{ Agrees.} \end{aligned} \quad (9)$$

Problem 2.

Use the “master formula” of chapter 23 of the notes to construct the representation of $SU(3)$ whose highest weight vector is twice the first fundamental weight vector. Draw the weight diagram. Then draw in this diagram the complex conjugate representation. What is the highest weight of the latter?

Solution: The first fundamental weight vector of $SU(3)$ is $\vec{\mu}_{def}^1 = (\frac{1}{2}, \frac{1}{2\sqrt{3}})$. So we begin with

$$\vec{\mu}_{HW} = (1, \frac{1}{\sqrt{3}}). \quad (10)$$

We use as simple roots of $SU(3)$

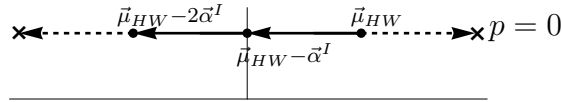
$$\vec{\alpha}^I = (1, 0); \quad \vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right). \quad (11)$$

For any weight $\vec{\mu}_R$ of a representation R the master formula reads

$$2 \frac{\vec{\alpha}^J \cdot \vec{\mu}_R}{\vec{\alpha}^J \cdot \vec{\alpha}^J} = -p + q, \quad \vec{\alpha}^J \text{ are simple roots.} \quad (12)$$

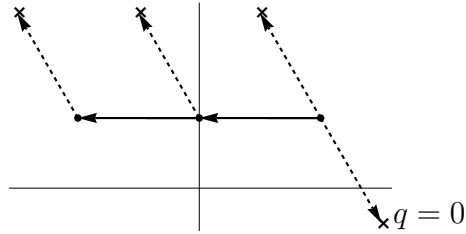
For the highest weight $p = 0$, so we begin with

Step I: $2 \frac{\vec{\alpha}^I \cdot \vec{\mu}_{HW}}{\vec{\alpha}^I \cdot \vec{\alpha}^I} = q = 2.$

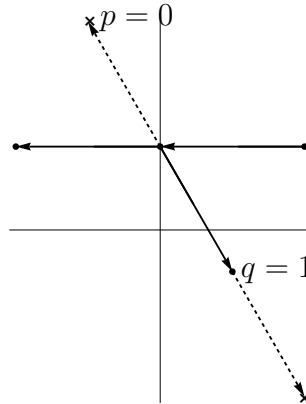


The notation $\cdots \rightarrow \times$ means that there is no weight at $\times$.

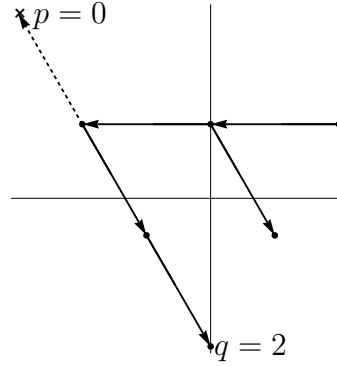
Step II: $2 \frac{\vec{\alpha}^{II} \cdot \vec{\mu}_{HW}}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = q = 0.$



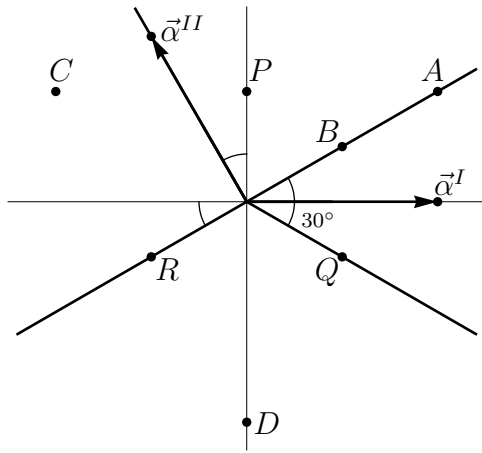
Step III: At $\vec{\mu}_{HW} - \vec{\alpha}^I$ we get $2 \frac{\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - \vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = -p + q = q = 1.$



Step IV: At $\vec{\mu}_{HW} - 2\vec{\alpha}^I$ we get $2\frac{\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - 2\vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = q = 2$.



We could repeat these steps at all remaining roots, but it is faster to use Weyl reflections. Weyl reflections show that we got all roots.



Weyl reflections:

$$A \rightarrow C \rightarrow D$$

$$P \rightarrow Q \rightarrow R$$

$$\vec{\alpha}^I = (1, 0)$$

$$\vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

$$B = \vec{\mu}_{def}^1 = \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right)$$

$$A = \left(1, \frac{1}{\sqrt{3}}\right) = 2\vec{\mu}_{def}^1$$

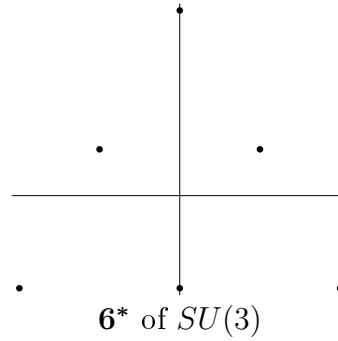
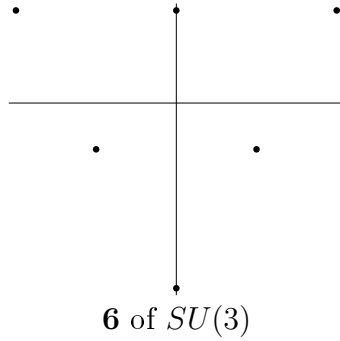
$$C = \left(-1, \frac{1}{\sqrt{3}}\right)$$

$$D = \left(0, -\frac{2}{\sqrt{3}}\right)$$

$$P = \left(0, \frac{1}{\sqrt{3}}\right)$$

$$Q = \left(\frac{1}{2}, -\frac{1}{2\sqrt{3}}\right)$$

$$R = \left(-\frac{1}{2}, -\frac{1}{2\sqrt{3}}\right)$$

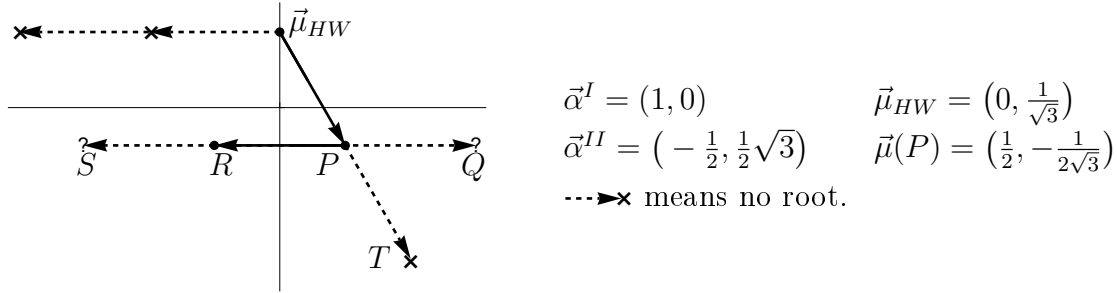


The complex conjugate diagram has as highest weight minus the lowest weight of the **6**, hence

$$\vec{\mu}_{HW}(\mathbf{6}^*) = \left(0, \frac{2}{\sqrt{3}}\right). \quad (13)$$

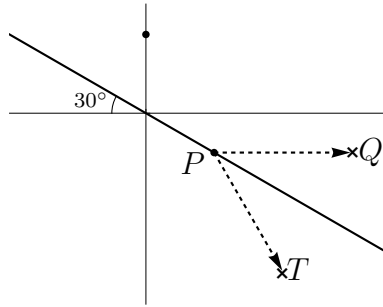
Bonus Problem.

We considered in class the $SU(3)$ representation whose highest weight is $(0, \frac{1}{\sqrt{3}})$. Using the master formula, we came as far as the figure indicates. We found a root at P , not at T .



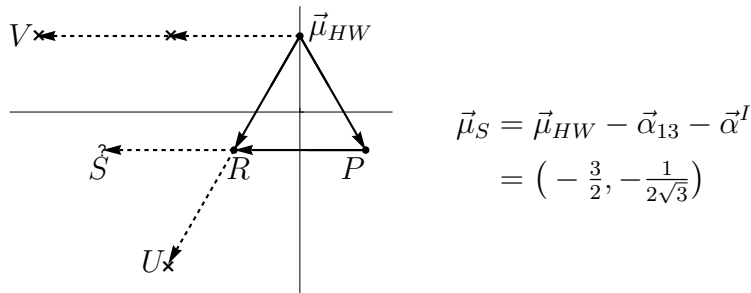
At P one finds from the master formula with $\vec{\alpha}^I$ that $-p + q = 1$. We want to show that $p = 0$ (no root at Q); then we only get a root at R . Prove, using the master formula, that there is no root at Q . Can one conclude this from Weyl reflections?

Solution: Under Weyl reflection Q is mapped to T , but there is no root at T , so there is no root at Q .



One can also prove this from the master equation. If there were a root at Q , there also would be a root at S . But using $\vec{\alpha}^{13}$ (the highest root) in the master formula at $\vec{\mu}_{HW}$ yields $q = 1$.

$$2 \frac{\vec{\alpha}^{13} \cdot \vec{\mu}_{HW}}{\vec{\alpha}^{13} \cdot \vec{\alpha}^{13}} = -p + q = q = 2\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \left(0, \frac{1}{\sqrt{3}}\right) = 1. \quad (14)$$



This implies that there a root at R but no root at U . Applying the master formula at S , we get

$$2 \frac{\vec{\alpha}^{II} \cdot \vec{\mu}_S}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = -p + q = 2\left(-\frac{3}{2}, -\frac{1}{2\sqrt{3}}\right) \left(-\frac{1}{2}, \frac{1}{2}\sqrt{3}\right) = 2\left(\frac{3}{4} - \frac{1}{4}\right) = 1. \quad (15)$$

But $p = 0$ (no root at V) and $q = 0$ (no root at U). Hence we get a contradiction: $1 = 0$, so there is no root at S .